

**Attacking Production Structure and Persistent Dominance
in the Premier League: Bottleneck Quality as the Dominant
Mechanism**

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Abstract

This paper examines why persistent performance inequalities emerge in the English Premier League despite a highly competitive labour market for talent, using player-level data from the 2015/16 to 2024/25 seasons. The central finding is that attacking bottleneck quality (measured as the 20th percentile of offensive quality within each squad) is the dominant and most robust predictor of points per game, both in isolation and in the presence of controls for total wage expenditure. Superstar quality, measured as the mean offensive contribution of the top three players, is positively associated with performance when considered alone but loses statistical significance once bottleneck quality is included in the model. This pattern is consistent with the view that distributional effects in team offensive production are concentrated in the lower tail of the squad rather than the upper tail. Labour market sorting evidence further shows that higher-quality offensive players are disproportionately concentrated at financially stronger clubs, reinforcing persistent performance gaps over time. The results are interpreted as robust associations rather than causal estimates.

Keywords: O-Ring theory, superstar theory, bottleneck quality, team production, Premier League, labour market sorting

1. Introduction

Persistent dominance is a defining feature of professional football. Despite the inherently competitive nature of league systems, a small number of clubs consistently dominate others over extended periods, capturing a disproportionate share of top league finishes and titles. This pattern is particularly evident in the English Premier League, where success has remained highly concentrated among a few top teams across the decade from 2015/16 to 2024/25. For example, Man City won the title in 6 out of 10 of these seasons. This raises a fundamental economic question: why do such persistent performance inequalities emerge in a setting where clubs compete in an open and highly fluid labour market for talent?

From an economic perspective, professional football provides a natural setting in which to study how talent is allocated across organisations and how team production functions operate when output depends on the coordinated efforts of many individuals. The existing literature has largely focused on the relationship between financial resources and team performance, with wage expenditure established as a strong predictor of league outcomes. However, this approach implicitly treats team production as principally additive, where overall performance is determined by cumulative talent or financial investment. It leaves open the question of whether the distribution of talent within teams (rather than its total level) plays a central role in shaping outcomes.

This paper contributes to that question by focusing specifically on attacking production structure. Given data constraints, the analysis measures player contributions through non-penalty expected goals (npxG), expected assists (xA), expected goal chain (xGChain), and expected goal buildup (xGBuildup), normalised within position-season groups. Defensive, off-ball, pressing, and tactical contributions are not captured by these metrics and are acknowledged as a limitation throughout. The paper is therefore best understood as an examination of how the distribution of offensive production capacity within squads relates to season-level performance, rather than a comprehensive model of team production.

The analysis for the paper introduces two distributional measures of squad composition: bottleneck quality, defined as the 20th percentile of offensive quality within each squad; and superstar quality, defined as the mean offensive quality of the top three players. These are used to test whether team performance is driven by upper-tail talent, constrained by lower-tail talent, or shaped by both simultaneously. Also, the paper examines whether labour market sorting reinforces performance differences over time, with past performance predicting subsequent squad quality.

The central empirical finding is that bottleneck quality is the dominant predictor of points per game. It is large in magnitude, highly significant across all specifications, and robust to alternative percentile cutoffs, alternative superstar definitions, and changes in the minimum playing-time threshold. Superstar quality is positively associated with performance when estimated in isolation, but loses statistical significance once bottleneck quality is included. This result is consistent with distributional effects in attacking production being concentrated in the lower tail of the squad rather than the upper tail, and it positions the paper in partial contrast to Szymanski and Wilkinson (2016), who find evidence more supportive of superstar theory than O-Ring theory using transfer fee proxies in an earlier sample period.

2. Literature Review

2.1 Superstar Theory

Rosen (1981) develops the theory of superstars, showing that in markets where output is scalable, and consumers favour the highest-quality performers, small differences in talent can generate disproportionately large differences in rewards. This leads to convex returns to skill, whereby marginal improvements in ability translate into large increases in income and market share. Adler (1985) extends this framework by highlighting the role of popularity and network effects, arguing that demand may concentrate around a small number of individuals even when differences in underlying talent are modest.

These mechanisms are particularly relevant in professional football, where elite players operate in a globally scalable market driven by broadcasting, sponsorship, and fan engagement. Empirical evidence confirms that football labour markets exhibit strong superstar characteristics, with player wages displaying convexity in performance measures such as goals and assists (Lucifora and Simmons, 2003). This supports the prediction that clubs may rationally concentrate resources on a small number of elite players in order to maximise both sporting and commercial returns.

However, the application of superstar theory to team production is not straightforward. While Rosen's framework focuses on individual output, football is inherently a multi-agent environment in which performance depends on coordinated interactions between players. Evidence from team-based sports shows that individual productivity is affected by peer effects and spillovers within teams (Arcidiacono et al., 2017), while football-specific studies suggest that the impact of talent may vary with the distribution of ability within the squad (Franck and Nüesch, 2010).

2.2 O-Ring Production Theory

Kremer (1993) introduces the O-Ring theory of production, in which output depends on the successful completion of multiple interdependent tasks. In such environments, complementarities imply that the productivity of any individual depends on the performance of others, creating a weak-link structure in which overall output is constrained by the least effective component. Even small deficiencies in one part of the production process can generate large reductions in total output.

This framework can naturally be applied to professional football, where match outcomes depend on coordinated actions across multiple players and positions. Defensive errors, misplaced passes, or weak pressing can undermine the contributions of stronger teammates. From this perspective, team performance may be characterised by complementarities in the O-Ring sense, where improving the quality of weaker squad members generates substantial gains.

The most directly relevant empirical test is Szymanski and Wilkinson (2016), who examine both O-Ring and superstar mechanisms using Premier League data from 1992/93 to 2013/14, measuring expected productivity through transfer fees. They find that the most expensive players exert the greatest impact on match results, while the least expensive players exert relatively little independent influence. Their results are more consistent with superstar theory than O-Ring theory, leading them to conclude that weak-link constraints do not appear to be the primary driver of Premier League performance when talent is proxied through transfer fees.

The present paper revisits this question using a different sample period, a different productivity proxy, and a different unit of analysis. Rather than game-level outcomes with transfer fee

values, this analysis uses season-level points per game and a process-based measure of offensive contributions. The finding that bottleneck quality is the dominant predictor in this period, and that it is robust across multiple specifications, suggests that distributional effects in offensive production may operate differently from what transfer fee proxies capture, possibly because the post-2015 period is characterised by much greater wage concentration and squad depth investment among top clubs, consistent with the growing talent and points concentration documented across European leagues during this era (Frick, Quansah and Lang, 2023).

2.3 Sorting in Labour Markets

A related strand of literature focuses on sorting in labour markets, where workers of similar ability match with organisations of similar productivity (Kremer and Maskin, 1996; Lazear, 1999). In environments characterised by complementarities, high-skill workers have incentives to join high-productivity firms where their marginal product is highest. This generates assortative matching and clustering of talent, reinforcing differences in productivity across organisations.

In professional football, these mechanisms imply that stronger clubs are more likely to attract higher-quality players due to their financial resources, institutional advantages, and competitive success. Empirical evidence supports this pattern, with higher-performing clubs typically employing stronger players and offering higher wages (Frick, 2007). More recent work shows that talent concentration contributes to competitive imbalance in European football, reinforcing persistent differences in performance across teams (Frick, Quansah and Lang, 2023).

2.4 Research Gap

Taken together, the existing literature suggests that team performance may depend simultaneously on elite talent, complementarities in production, and labour market sorting. However, empirical work has largely relied on aggregate measures of team quality, such as wage expenditure or average player ability, which do not distinguish between teams with balanced squads and those relying on a small number of elite players. This paper addresses this gap by constructing distributional measures that separately identify upper- and lower-tail offensive quality, testing their independent contributions to performance, and assessing whether sorting dynamics reinforce performance gaps over time.

3. Data and Empirical Strategy

3.1 Data Sources and Sample Construction

The analysis combines three publicly available datasets to construct a panel of player and club-level observations for the English Premier League covering the 2015/16 to 2024/25 seasons. Team performance data are obtained from FBref, which provides season-level information on league points and match outcomes. Player performance data are sourced from Understat, which reports advanced metrics (definitions table included in Appendix), including non-penalty expected goals (npxG), expected assists (xA), expected goal chain (xGChain), and expected goal buildup (xGBuildup). Club wage data are drawn from Capology, which provides estimates of individual player salaries and total club wage expenditure.

The initial dataset contains 5,353 player-season observations and 200 club-season observations. Goalkeepers are excluded from the quality measure construction, as their offensive contributions as measured by these metrics are negligible and the resulting index would be uninformative for this position group. The player sample is further restricted to

outfield players with at least 900 minutes of playing time in a given season. This restriction reduces noise from small sample sizes and ensures that performance metrics reflect sustained contributions rather than sporadic appearances. The 900-minute threshold is a standard choice in the sports analytics literature; a robustness check using a 450-minute threshold is reported in Section 4.3 and produces consistent results. After cleaning and merging across data sources, the final player-level dataset contains 2,937 observations across three outfield position groups.

At the club level, merging player-level quality aggregates with club wage and performance data yields a panel of 200 club-season observations. Requiring complete information on all variables used in the main regression models reduces the final estimation sample to 136 club-season observations. This reduction primarily reflects incomplete wage data for newly promoted clubs, which are less frequently covered by Capology. The estimation sample, therefore, skews toward established Premier League sides. If anything, this makes the analysis more conservative: established clubs have more stable squad structures, and their offensive metrics are more precisely estimated over a full season.

3.2 Constructing a Bounded Measure of Offensive Quality

A central challenge is to construct a measure of player quality that is comparable across teams, positions, and seasons. Rather than relying on raw totals, the analysis uses per-90-minute rates of four process-based offensive metrics: npxG, xA, xGChain, and xGBuildup. Each metric is standardised within position-season groups to produce a z-score, and the four z-scores are averaged to form a composite offensive production measure. The equal weighting of components reflects the absence of a theoretical basis for differential weighting; the within-position-season standardisation ensures each metric contributes comparably to the composite regardless of its raw scale.

It should be noted explicitly that this framework captures offensive production involvement and cannot reflect defensive contributions, off-ball movement, aerial duels won, pressing intensity, or tactical positioning. No publicly available dataset currently provides reliable per-player metrics for these dimensions at the scale required for a ten-season panel analysis. This is a recognised constraint of the sports analytics literature more broadly, not specific to this study. As a result, the analysis is best understood as an examination of how the distribution of attacking production capacity within squads relates to season performance, and all references to 'player quality' or 'squad quality' in what follows should be read as shorthand for offensive quality as defined above.

To ensure comparability across seasons and reduce the influence of extreme values, the composite measure is benchmarked against a position-specific replacement level defined as the 20th percentile of offensive production among regular players within each position group. The replacement-relative measure is then normalised within each season-position group and transformed using a logistic function to produce a bounded quality index q lying between zero and one. This transformation preserves relative differences in performance while preventing outliers from disproportionately influencing the analysis.

3.3 Aggregating to the Club Level

Player-level quality measures are aggregated to the club-season level to construct distributional indicators of squad offensive composition. Three primary measures are constructed. Average squad quality (avg_q) captures the overall level of offensive ability. Bottleneck quality is measured using the 20th percentile of player quality within each squad (p20_q), serving as the primary indicator of weak-link constraints in offensive production. Superstar quality is

captured by the mean quality of the top three players in each squad (top3_q), reflecting the concentration of elite offensive talent.

Alternative measures (including the 10th percentile, minimum quality, top-1, and top-5 superstar definitions) are constructed for robustness checks. All measures are standardised within season prior to estimation, so coefficients can be interpreted as the effect of a one-standard-deviation increase relative to the seasonal distribution.

3.4 Wage Regressions as Validation

Before examining team-level performance, the analysis evaluates whether the constructed offensive quality measure is priced in the labour market. If the measure captures economically meaningful productivity, it should be positively associated with player wages. Player-level wage regressions relate log annual wages to the quality measure q , controlling for position, age, season effects, and club wage environment. As reported in Table 2, higher offensive quality is strongly and significantly associated with higher wages across all specifications, providing evidence that the constructed measure reflects dimensions of player performance valued by clubs.

The analysis also estimates a sorting regression relating player quality to the log club wage bill, controlling for position and season fixed effects. This tests whether higher-quality offensive players are disproportionately concentrated at financially stronger clubs, consistent with positive assortative matching. The sorting regression yields a positive and significant coefficient on the club wage bill (coefficient 0.024, clustered SE 0.004, $p < 0.001$), indicating that richer clubs systematically employ higher-quality offensive players. This is consistent with labour market sorting reinforcing performance gaps between clubs over time.

3.5 Club-Level Performance Models

The main empirical analysis estimates a sequence of club-level regression models relating points per game (PPG_{it}) to financial resources and the distribution of offensive squad quality. The baseline model includes only log wage expenditure and season fixed effects. Subsequent specifications introduce average squad quality, bottleneck quality, and superstar quality sequentially and jointly, allowing the analysis to assess which distributional dimensions matter independently. A persistence model further introduces lagged points per game and lagged wage expenditure to examine whether current performance reflects dynamic carry-over effects.

Season fixed effects are included throughout to control for common time shocks. Club fixed effects are not included in the primary specifications. Omitting them is a deliberate choice: with 136 observations across approximately 20 clubs, club fixed effects would absorb the cross-sectional variation in squad structure that is central to the paper's contribution. The interesting variation is structural (why Manchester City and Liverpool consistently outperform Bournemouth and Brentford), and fixed effects would eliminate this. A club fixed effects robustness check is nonetheless estimated and reported in Table 3, column (6). All standard errors are clustered at the club level.

The empirical strategy is designed to identify theoretically meaningful associations between offensive squad structure and performance. Team performance, wage expenditure, and squad quality are jointly determined, raising the possibility of simultaneity and omitted-variable bias. The results throughout are interpreted as robust associations consistent with the theoretical framework, rather than definitive causal estimates.

4. Results

4.1 Descriptive Statistics

Table 1: Descriptive Statistics - Club-Season Panel

Variable	Mean	Std. Dev.	Min	Median	Max	N
Points per Game	1.405	0.503	0.316	1.289	2.632	136
Log Wage Bill	18.239	0.540	16.992	18.210	19.293	136
Avg. Offensive Quality	0.499	0.158	0.241	0.443	0.890	136
Bottleneck Quality	0.392	0.159	0.162	0.331	0.856	136
Superstar Quality	0.686	0.167	0.344	0.685	0.980	136

Notes: 136 club-season observations. All quality measures are bounded between 0 and 1. Bottleneck quality is the 20th percentile of offensive quality within each squad. Superstar quality is the mean of the top three players.

Points per game ranges from 0.32 to 2.63, representing the full spectrum from relegated clubs to league champions. The standard deviation of 0.50 means that typical clubs sit roughly half a point per game away from the sample mean, which over a 38-game season translates to approximately 19 points. That is substantial inequality in outcomes, and explaining where it comes from is what this paper is about.

The wage bill variation is even starker. Log wage bills range from 16.99 to 19.29, which in raw terms corresponds to roughly €22m at the low end and €267m at the high end, a sevenfold difference in annual expenditure. This financial inequality is the starting point for existing explanations of competitive imbalance. The question this paper asks is whether that financial story is the whole story, or whether the distribution of talent within squads adds something beyond it.

The quality measures reveal an important pattern. Bottleneck quality has a mean of 0.39 and a standard deviation of 0.16. Superstar quality has a mean of 0.69 and a standard deviation of 0.17. The near-identical standard deviations are worth noting: clubs are roughly as heterogeneous in the quality of their weakest offensive contributors as in the quality of their best. If performance variation were driven primarily by differences in elite talent, one would expect superstar quality to be the more dispersed of the two. The fact that it is not provides early descriptive evidence that the lower tail of the squad distribution may be where competitive differences are concentrated.

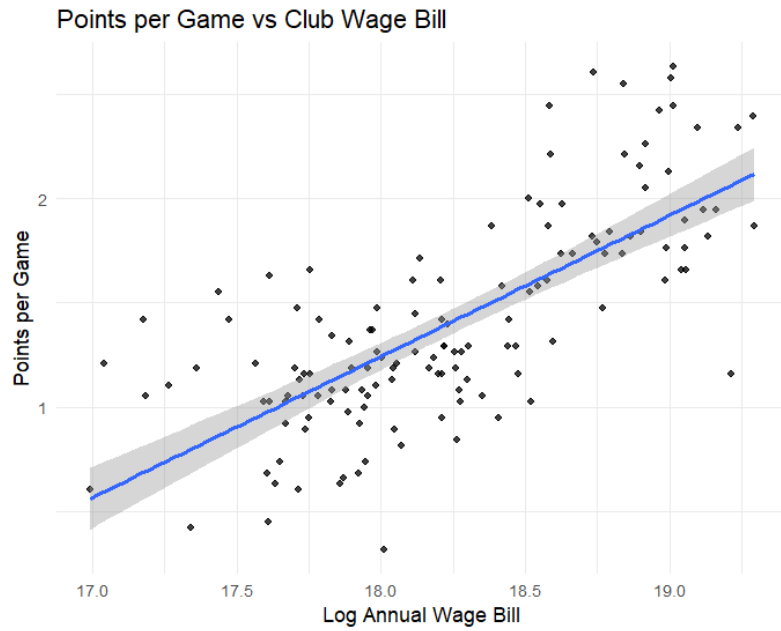


Figure 1 shows a clear positive association between the 20th percentile of squad offensive quality and season points per game, consistent with a weak-link interpretation of team performance.

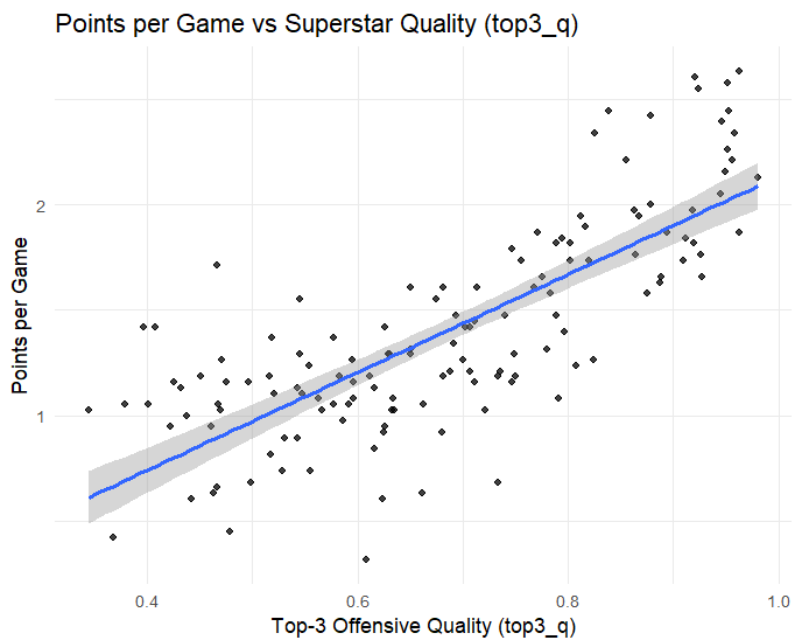


Figure 2 illustrates the well-established positive relationship between club wage expenditure and league outcomes, which serves as the financial benchmark for the regression analysis.

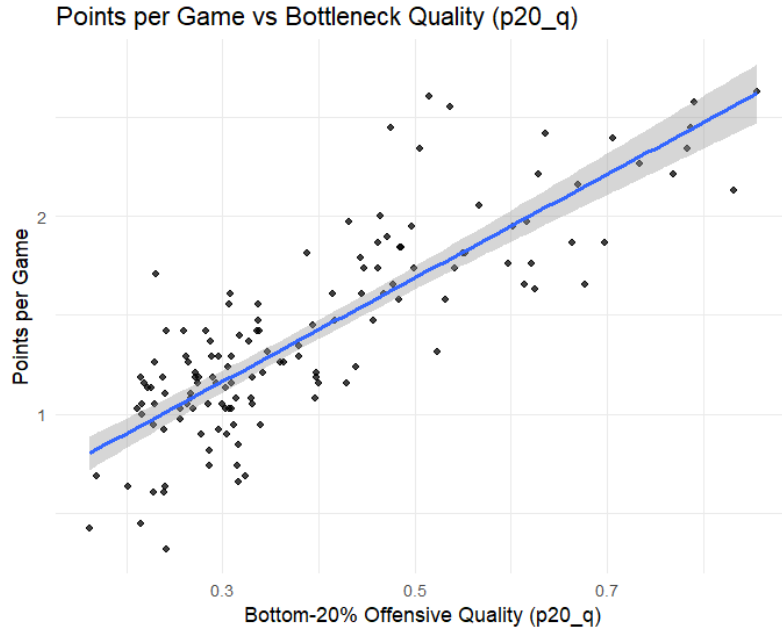


Figure 3 shows a positive association between the mean quality of the top three players and points per game, consistent with superstar theory, though the association is somewhat weaker than that observed for bottleneck quality.

4.2 Wage Pricing and Sorting

Before examining team-level performance, it is necessary to establish that the quality measure q captures economically meaningful variation. If the measure reflected noise rather than genuine productivity, there would be no theoretical basis for using it to explain team outcomes. The wage regressions in Table 2 provide this validation.

In the baseline specification (A1), the quality coefficient is 0.748 (SE 0.094, $p < 0.01$), indicating that moving from replacement level to elite offensive contribution is associated with wages approximately 75 per cent higher. This is a large return to quality and confirms that clubs price the measure substantially in the labour market. Adding team fixed effects (A2) reduces the coefficient to 0.506 (SE 0.083, $p < 0.01$), as expected: within-club wage structures are compressed by contract timing and squad harmony considerations, so the within-club return to quality is somewhat smaller than the cross-sectional return. The player fixed effects specification (A3) reduces the coefficient further to 0.047 (SE 0.090), which is no longer significant. This attenuation is expected and does not reflect a limitation of the measure; player quality scores do not vary substantially within individuals across seasons, leaving little within-player variation for the estimator to identify. The A1 and A2 results, which are the relevant validation tests, confirm that offensive quality is strongly priced in the labour market.

There is also evidence that quality is priced more steeply at richer clubs. The interaction between quality and club wage environment reaches significance in A2 (0.301, SE 0.133, $p < 0.05$), consistent with the idea that financially stronger clubs pay a larger premium for high-quality players. This result is not robust across all error structures and is therefore interpreted cautiously.

The sorting regression relates individual player quality to the club log wage bill, controlling for position and season fixed effects. The coefficient is 0.024 (SE 0.004, $p < 0.001$), indicating

that richer clubs systematically employ higher-quality offensive players throughout their squads, not merely at the top of the distribution. This positive assortative matching result motivates the dynamic analysis in Section 4.5: if financial strength raises offensive quality across the entire squad distribution, financial dominance may reinforce itself over time, specifically through lower-tail squad quality rather than solely through the acquisition of elite players.

Table 2: Player Wage Pricing Models

	Baseline (A1)	Team FE (A2)	Player FE (A3)
Offensive Quality (q, centred)	0.748*** (0.094)	0.506*** (0.065)	0.047 (0.066)
Club Wage Env. (centred)	0.694*** (0.032)	0.490*** (0.062)	0.373*** (0.027)
Quality $\times$ Wage Env.	0.136 (0.122)	0.232** (0.086)	—
Position FE	Yes	Yes	—
Season FE	Yes	Yes	Yes
Team FE	No	Yes	—
Player FE	No	No	Yes
N	2,937	2,937	2,468
R ²	0.517	0.579	0.847

*Notes: Dependent variable is log annual wage. Standard errors clustered by player are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Age squared included in all specifications. Season fixed effects included throughout. Offensive quality (q) is a bounded measure of attacking production constructed from $npxG$, xA , $xGChain$, and $xGBuildup$, normalised within position-season groups.*

4.3 Main Results: Bottleneck Quality as the Dominant Mechanism

Table 3 presents the central findings. The models are estimated sequentially so that the contribution of each component (financial resources, average quality, bottleneck quality, superstar quality) can be assessed in isolation first and then in direct comparison. The key question is not whether these mechanisms are individually significant, but which retains explanatory power when placed alongside the others.

Model (1) establishes the financial benchmark. The log wage bill coefficient is 0.686 (SE 0.106, $p < 0.01$), with $R^2 = 0.541$, replicating the standard result in the sports economics literature. Financial resources explain approximately 54 per cent of the variation in points per game.

Model (2) introduces average offensive squad quality as an aggregate benchmark. Average quality enters at 0.380 (SE 0.041, $p < 0.01$), and its inclusion diminishes the wage coefficient by nearly 80 per cent, from 0.686 to 0.139. The R^2 rises strongly to 0.630. This pattern points out that the association between wage expenditure and performance operates primarily through the offensive quality of the players those wages secure. Once the composition of the squad is measured directly, the wage bill loses most of its independent predictive power. The high R^2 of this specification also establishes a demanding benchmark for the distributional models that follow: average quality, as a summary of the full squad distribution, will naturally achieve high

overall fit. The contribution of the distributional approach in Models (3) through (5) is not to surpass this fit, but to identify which part of the squad distribution accounts for it.

Model (3) replaces average quality with bottleneck quality (the 20th percentile of offensive quality within each squad) to test the weak-link mechanism in isolation. The coefficient is 0.316 (SE 0.031, $p < 0.01$), with $R^2 = 0.697$. In practical terms, a one-standard-deviation improvement in lower-tail offensive quality is associated with approximately 13 additional points over a full season, roughly the margin between a mid-table finish and a top-four contention. This is consistent with O-Ring production: clubs whose weakest regular offensive contributors are stronger relative to the seasonal distribution accumulate substantially more points, independent of their wage expenditure.

Model (4) replaces bottleneck quality with superstar quality (the mean quality of the top 3 players) to test the superstar mechanism in isolation. The coefficient is 0.285 (SE 0.059, $p < 0.01$), with $R^2 = 0.680$. Upper-tail offensive quality is also a significant predictor when taken in isolation, consistent with superstar theory. A one-standard-deviation improvement in elite offensive talent is associated with almost 11 additional points per season.

The decisive test is Model (5), which includes both measures simultaneously and allows the data to arbitrate between the two theories. Bottleneck quality retains a large and precisely estimated coefficient of 0.292 (SE 0.050, $p < 0.01$). Superstar quality falls to 0.078 (SE 0.071, $p = 0.270$) and is no longer statistically distinguishable from zero. The predictive content of superstar offensive quality is not independent of bottleneck quality in the joint specification. The high correlation between the two measures ($r = 0.852$) means that clubs with strong elite players also tend to have stronger weaker players. What appeared as a superstar effect in Model (4) was substantially driven by the association between elite talent and squad depth. The performance advantage belongs to the lower tail.

This result partially contrasts with Szymanski and Wilkinson (2016), who find evidence more consistent with superstar theory over an earlier sample period using transfer fees as a talent proxy. Two interpretations are plausible. Transfer fees and process-based offensive metrics may weigh different dimensions of player contribution, with fees reflecting market-assessed value across all roles and the present metrics capturing offensive production involvement specifically. Additionally, the post-2015 era has seen substantial investment in squad depth among top clubs, and consistent offensive quality throughout the squad may represent a more binding performance constraint in the current period than it was previously.

Model (6) adds club fixed effects to the joint specification, controlling for all time-invariant club characteristics (historical prestige, managerial culture, fan base) that could simultaneously predict squad composition and performance. Bottleneck quality remains positive and significant at 0.261 (SE 0.068, $p < 0.01$); superstar quality remains insignificant at 0.062 (SE 0.058). The wage coefficient loses significance, as expected given that wages are highly persistent and their cross-sectional variation is absorbed by the club fixed effect. The survival of the bottleneck result under this demanding specification indicates that the association is not simply an artefact of certain clubs being historically strong across all dimensions simultaneously.

Table 3: Main Regression Results -- Bottleneck vs. Superstar

	(1) Wages	(2) Avg Q	(3) Bottleneck	(4) Superstar	(5) Both	(6) Both+FE
Log Wage Bill	0.686*** (0.058)	0.412*** (0.063)	0.286*** (0.074)	0.296*** (0.075)	0.169* (0.069)	0.167 (0.191)
Avg. Offensive Quality		0.245*** (0.059)				
Bottleneck Quality (p20_q)			0.316*** (0.030)		0.292*** (0.050)	0.261*** (0.068)
Superstar Quality (top3_q)				0.285*** (0.059)	0.078 (0.071)	0.062 (0.058)
Season FE	Yes	Yes	Yes	Yes	Yes	Yes
Club FE	No	No	No	No	No	Yes
N	136	136	136	136	136	136
R ²	0.541	0.630	0.697	0.680	0.714	0.762

Notes: Dependent variable is points per game. Standard errors clustered by club are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All quality measures are standardised within season. Season fixed effects included in all models. Model (6) additionally includes club fixed effects.

4.4 Robustness Checks

Table 4 addresses the two modelling choices that could arguably be questioned: the definition of bottleneck quality as the 20th percentile, and the restriction of the player sample to those with at least 900 minutes of playing time.

To address the bottleneck definition, we perform a robustness check with an alternative cutoff at the 10th percentile (0.326, SE 0.030, $p < 0.01$) and the minimum player quality (0.244, SE 0.031, $p < 0.01$), both of which produce large, precisely estimated, and highly significant coefficients. The result is stable across the lower tail of the distribution and is not an artefact of where the cutoff is drawn. For superstar quality, alternative definitions using the single best player (0.213, SE 0.054, $p < 0.01$) and the top five players (0.334, SE 0.053, $p < 0.01$) confirm that upper-tail quality is significant in isolation across all specifications, consistent with Model (4). These alternatives are not tested jointly with bottleneck quality because the pattern of superstar attenuation established in Table 3 is consistent across all superstar definitions.

The 450-minute threshold check addresses the concern that the 900-minute restriction excludes fringe players who may represent genuine bottlenecks. Rebuilding the quality measure with the lower threshold and re-estimating produces a bottleneck coefficient of 0.340 (SE 0.032, $p < 0.01$), virtually identical to the main result of 0.316. The bottleneck result is not an artefact of restricting attention to established regulars; including lower-minute players in the quality distribution leaves the coefficient essentially unchanged.

Table 4: Robustness Checks

	p10 Bottleneck	Min Bottleneck	Top-1 Superstar	Top-5 Superstar	450-min
Log Wage Bill	0.232** (0.085)	0.366*** (0.102)	0.417*** (0.087)	0.214** (0.067)	0.215** (0.082)
Quality Measure	0.326*** (0.030)	0.244*** (0.031)	0.213*** (0.054)	0.334*** (0.053)	0.340*** (0.032)
Season FE	Yes	Yes	Yes	Yes	Yes
N	136	136	136	136	136

*Notes: Each column reports a separate regression of points per game on log wage bill and a single quality measure, with season fixed effects. Standard errors clustered by club in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Columns (1)–(2) use alternative bottleneck definitions; (3)–(4) use alternative superstar definitions; (5) uses a 450-minute playing-time threshold to construct the bottleneck measure.*

4.5 Persistence and Sorting

Table 5 examines whether the bottleneck mechanism helps explain why dominance in the Premier League persists over time. The models are estimated on a lagged panel of 116 club-season observations, adding prior-season variables sequentially.

Model (E1) establishes the baseline persistence result: lagged points per game has a coefficient of 0.808 (SE 0.062, $p < 0.01$), indicating that approximately 80 per cent of a club's performance advantage in one season carries forward to the next. The Premier League is a highly persistent competition.

Model (E2) adds lagged wage expenditure. The lagged wage coefficient is 0.343 (SE 0.077, $p < 0.01$), and the persistence coefficient falls to 0.540. Sustained financial investment accounts for a meaningful share of the serial correlation in performance, but a persistence coefficient of 0.540 indicates substantial continuity beyond financial resources alone.

Model (E3) adds current-season bottleneck quality, and the most informative result in this section emerges. The bottleneck coefficient is 0.298 (SE 0.036, $p < 0.01$), and the persistence coefficient falls sharply from 0.540 to 0.222. The lagged wage coefficient simultaneously declines and loses significance. Much of what appeared as performance persistence is explained by the consistency of lower-tail offensive squad quality across seasons. Clubs that sustain strong bottleneck quality from one season to the next are precisely those that appear persistently dominant; the serial correlation in points is substantially a serial correlation in squad depth.

Model (E4) adds superstar quality alongside the bottleneck. The pattern from Table 3 replicates exactly: bottleneck quality remains significant at 0.254 (SE 0.057, $p < 0.01$), superstar quality is insignificant at 0.067 (SE 0.069), and the persistence coefficient changes only marginally from Model (E3). Superstar quality adds nothing to the dynamic explanation once the bottleneck is controlled.

The sorting regression connects the dynamic findings back to the labour market. A one-unit increase in the log club wage bill is associated with a 0.024-unit increase in individual player offensive quality (SE 0.004, $p < 0.001$), controlling for position and season. Richer clubs attract

higher-quality offensive players throughout their squads, including at the lower tail. The mechanism linking financial advantage to sustained sporting dominance, therefore, appears to operate primarily through squad depth: financial resources are associated with stronger bottleneck quality, which in turn correlates with higher performance, and performance reinforces the club's capacity to attract talent in the following season. This dynamic pattern, where financial advantage is associated with greater squad depth, which in turn correlates with higher points accumulation, potentially reinforcing future financial capacity, provides a coherent account of why competitive inequality in the Premier League is both substantial and persistent.

Table 5: Persistence and Competitive Balance Models

	(E1) Persistence	(E2) +Wages	(E3) +Bottleneck	(E4) +Both
Lagged Points per Game	0.808*** (0.062)	0.540*** (0.078)	0.222** (0.069)	0.210** (0.066)
Lagged Log Wage Bill		0.343*** (0.077)	0.144 (0.075)	0.122 (0.074)
Bottleneck Quality (p20_q)			0.298*** (0.036)	0.254*** (0.057)
Superstar Quality (top3_q)				0.067 (0.069)
Season FE	Yes	Yes	Yes	Yes
N	116	116	116	116
R ²	0.665	0.715	0.793	0.797

*Notes: Dependent variable is points per game. Lagged variables refer to the prior season. Standard errors clustered by club in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All quality measures standardised within season. Season fixed effects included in all models.*

5. Conclusion

This paper examined whether the distribution of offensive production capacity within squads, rather than aggregate financial resources alone, helps explain persistent performance differences across Premier League clubs. Using a bounded measure of offensive quality constructed from four process-based metrics across the 2015/16 to 2024/25 seasons, the analysis estimated a sequence of performance models designed to distinguish between upper-tail superstar effects and lower-tail bottleneck effects.

The central finding is that the attacking bottleneck quality is the dominant and most robust predictor of points per game. It is highly significant across all specifications, robust to alternative definitions of the bottleneck, robust to relaxing the playing-time threshold, and survives the inclusion of club fixed effects. Superstar quality is positively associated with performance in isolation but is not robustly independent once bottleneck quality is controlled for. This result suggests that the distribution of offensive production in the Premier League is best characterised as lower-tail constrained rather than upper-tail driven. The results partially

contrast with Szymanski and Wilkinson (2016), who find more support for superstar theory in an earlier period using transfer fees, and suggest that distributional effects in the post-2015 era may operate differently, possibly reflecting the greater investment in squad depth among competitive clubs in recent seasons.

The persistence and sorting results add a dynamic dimension to these findings. Lagged performance is strongly associated with current performance, but this serial correlation is substantially reduced when current offensive squad structure is controlled for. Labour market sorting further shows that financially stronger clubs attract higher offensive quality throughout the squad distribution, implying that financial dominance translates into sporting dominance largely through its effect on distributional offensive quality rather than through aggregate investment alone.

Several limitations should be acknowledged. The quality measure is based on offensive production metrics and cannot capture defensive contributions, off-ball movement, aerial duels, pressing intensity, or tactical positioning. No publicly available dataset currently provides reliable per-player metrics for these dimensions at the scale required for this analysis. Future work incorporating possession value models, SPADL-type action data, or tracking-based metrics capturing defensive and pressing contributions would allow weak-link constraints to be tested across both sides of the game rather than only just the attacking dimension alone. The analysis is observational, and the results are subject to the standard concerns about simultaneity and omitted variable bias. They should be read as robust associations, not causal estimates.

More broadly, these findings suggest that persistent dominance in professional football arises not simply from how much talent clubs possess, but from how that talent is distributed within squads. Understanding performance in team-based environments requires moving beyond aggregate measures of quality and attending to the full distribution of contributions within the production unit.

6. Further Analysis

Beyond the core results reported above, exploratory analysis examined several additional mechanisms. These include tournament theory tests assessing whether internal wage inequality within clubs is associated with stronger performance; competitive balance and persistence models examining whether financial and talent concentration reinforce dominance over time; and the role of UEFA competition participation as a potential reinforcement channel for persistent league performance. Wage inequality measures are not independently associated with performance once squad quality is controlled for, consistent with the interpretation that internal wage dispersion is endogenous to squad composition rather than an independent incentive mechanism. The competitive balance and European competition results broadly support the main findings but are omitted from the primary results to preserve narrative focus within the word limit. Full results are available on request.

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Appendix

Table 6: Correlation Matrix - Main Variables

Variable	PPG	Log Wage	Avg Q	p20 Q	Top3 Q	Europe
PPG	1.000	0.726	0.851	0.830	0.773	0.676
Log Wage	0.726	1.000	0.771	0.757	0.711	0.741
Avg Q	0.851	0.771	1.000	0.974	0.938	0.761
p20 Q	0.830	0.757	0.974	1.000	0.852	0.745
Top3 Q	0.773	0.711	0.938	0.852	1.000	0.682
Europe	0.676	0.741	0.761	0.745	0.682	1.000

Notes: Pairwise correlations across 136 club-season observations.

Variable Definitions

Variable	Definition	Source
Points per Game (PPG)	Total league points divided by 38 matches	FBref
Log Wage Bill	Natural logarithm of total annual club wage expenditure in euros	Capology
np _{xG}	Non-penalty expected goals per 90 minutes	Understat
x _A	Expected assists per 90 minutes	Understat
x _{GChain}	Expected goals from all possessions the player was involved in, per 90 minutes	Understat
x _{GBuildup}	Expected goals from all possessions excluding shots and key passes, per 90 minutes	Understat
Offensive Quality (q)	Bounded quality index constructed by averaging within-position-season z-scores of np _{xG} , x _A , x _{GChain} , and x _{GBuildup} per 90, benchmarked against position-specific replacement level, and transformed using a logistic function. Ranges between 0 and 1	Understat
Replacement Level	20th percentile of offensive production among regular outfield players within each position group and season	Derived
Bottleneck Quality (p _{20_q})	20th percentile of offensive quality (q) within each club-season squad	Derived
Superstar Quality (top _{3_q})	Mean offensive quality (q) of the top three players within each club-season squad	Derived
Average Squad Quality (avg_q)	Mean offensive quality (q) across all qualifying outfield players within each club-season squad	Derived
p _{10_q}	10th percentile of offensive quality within each club-season squad. Used as robustness check alternative to p _{20_q}	Derived
min_q	Minimum offensive quality within each club-season squad. Used as robustness check	Derived
top _{1_q}	Maximum offensive quality within each club-season squad. Used as robustness check alternative to top _{3_q}	Derived
top _{5_q}	Mean offensive quality of the top five players within each club-season squad. Used as robustness check	Derived
Regular Player	Outfield player with at least 900 minutes of playing time in a given season	Derived
UCL	Binary indicator equal to 1 if the club participated in the UEFA Champions League during that season	Manual

UEL	Binary indicator equal to 1 if the club participated in the UEFA Europa League during that season	Manual
UECL	Binary indicator equal to 1 if the club participated in the UEFA Europa Conference League during that season	Manual
Europe (any)	Binary indicator equal to 1 if the club participated in any UEFA club competition during that season	Derived
Lagged PPG	Points per game in the prior season	Derived
Lagged Log Wage Bill	Natural logarithm of total wage expenditure in the prior season	Derived

Data Sources

There are three sources. FBref (fbref.com) provides season-level team performance data including total points, wins, draws, losses, and goal difference for each Premier League club across the 2015/16 to 2024/25 seasons. Understat (understat.com) provides player-level advanced metrics, including non-penalty expected goals (npG), expected assists (xA), expected goal chain (xGChain), and expected goal buildup (xGBuildup), reported on a per-appearance basis. Capology (capology.com) provides estimates of individual player annual wages in euros and total club wage expenditure, sourced from publicly reported contracts and reliable secondary sources.